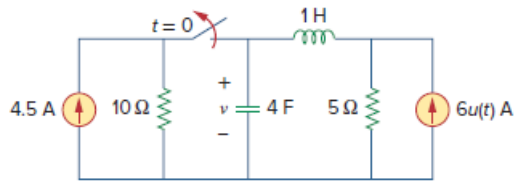


Problem

Find $v(t)$ for $t > 0$ in the circuit in Fig. 16.50.

Figure 16.50:



Step-by-step solution

Step 1 of 8

Refer to Figure 16.50 in the text book.

At $t < 0$, inductor acts as a short circuit and capacitor acts as an open circuit.

For $t < 0$,

$$u(t) = 0$$

Therefore, the current source of $6u(t)$ is zero.

Draw the circuit diagram for $t < 0$ as shown in Figure 1.

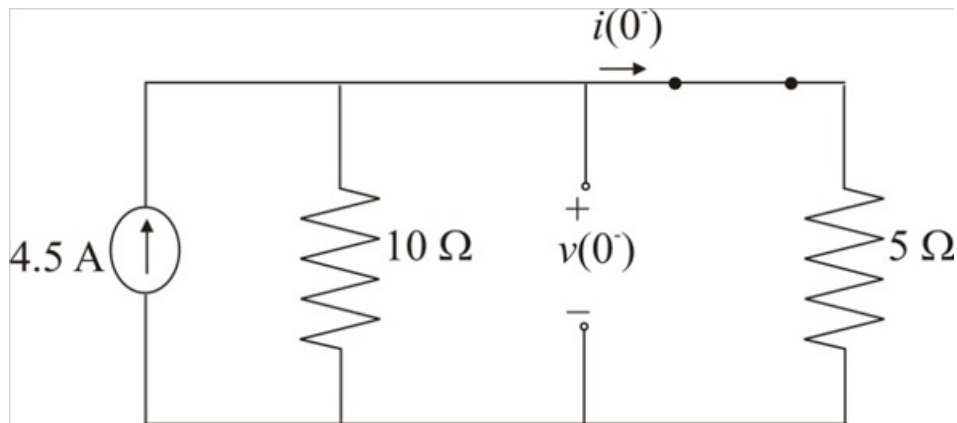


Figure 1

Step 2 of 8

Calculate the value of the current $i(0^-)$ using current division rule.

$$\begin{aligned} i(0^-) &= (4.5) \left(\frac{10}{10+5} \right) \\ &= (4.5) \left(\frac{10}{15} \right) \\ &= 3 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 3 A.

Step 3 of 8

Calculate the value of the voltage $v(0^-)$.

$$v(0^-) = 5i(0^-)$$

Substitute 3 A for $i(0^-)$.

$$\begin{aligned} v(0^-) &= (5)(3) \\ &= 15 \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(0^-)$ is 15 V .

Step 4 of 8

For $t > 0$,

$$u(t) = 1$$

Therefore, the current source of $6u(t)$ is 6 V .

Represent the circuit parameters in Laplace transform as shown in Figure 2.

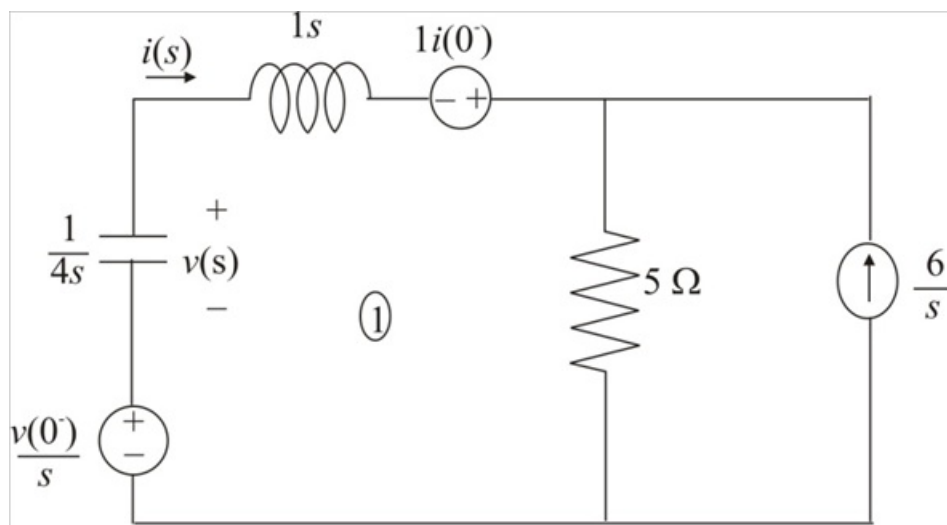


Figure 2

Step 5 of 8

Apply Kirchhoff's Voltage Law in loop-1.

$$i(s)\left(s + \frac{1}{4s}\right) + 5\left(i(s) + \frac{6}{s}\right) - i(0^-) - \frac{v(0^-)}{s} = 0$$

Substitute 15 V for $v(0^-)$ and 3 A for $i(0^-)$.

$$i(s)\left(s + \frac{1}{4s}\right) + 5i(s) + \frac{30}{s} - 3 - \frac{15}{s} = 0$$

$$i(s)\left(s + \frac{1}{4s} + 5\right) - 3 + \frac{15}{s} = 0$$

$$i(s)\left(s + \frac{1}{4s} + 5\right) = 3 - \frac{15}{s}$$

$$i(s)\left(\frac{4s^2 + 20s + 1}{4s}\right) = \frac{3s - 15}{s}$$

$$i(s)(4s^2 + 20s + 1) = 12s - 60$$

$$i(s) = \frac{12s - 60}{4s^2 + 20s + 1}$$

Therefore, the value of the current $i(s)$ is $\frac{12s - 60}{4s^2 + 20s + 1}$.

Step 6 of 8

Calculate the value of the voltage across the capacitor.

$$V(s) = v(s) + \frac{v(0^-)}{s}$$

Substitute $-\left(\frac{1}{4s}\right)i(s)$ for $v(s)$ and 15 V for $v(0^-)$.

$$V(s) = -\left(\frac{1}{4s}\right)i(s) + \frac{15}{s}$$

Substitute $\frac{12s - 60}{4s^2 + 20s + 1}$ for $i(s)$.

$$\begin{aligned} V(s) &= -\left(\frac{1}{4s}\right)\left(\frac{12s - 60}{4s^2 + 20s + 1}\right) + \frac{15}{s} \\ &= \frac{15 - 3s}{s(4s^2 + 20s + 1)} + \frac{15}{s} \end{aligned}$$

Step 7 of 8

Convert $\frac{15 - 3s}{s(4s^2 + 20s + 1)}$ into its partial fraction form.

$$\begin{aligned} \frac{15 - 3s}{s(4s^2 + 20s + 1)} &= \frac{A}{s} + \frac{Bs + C}{4s^2 + 20s + 1} \\ &= \frac{A(4s^2 + 20s + 1) + s(Bs + C)}{s(4s^2 + 20s + 1)} \end{aligned}$$

Compare the coefficients of s^2 , s and the constant terms on both the sides.

$$4A + B = 0 \dots\dots (1)$$

$$20A + C = -3 \dots\dots (2)$$

$$A = 15$$

Step 8 of 8

Substitute the value of A in equation (1) and (2).

$$B = -60$$

$$\begin{aligned} C &= -3 - 20(15) \\ &= -303 \end{aligned}$$

Recall the expression for $V(s)$.

$$\begin{aligned} V(s) &= \frac{15-3s}{s(4s^2+20s+1)} + \frac{15}{s} \\ &= \frac{A}{s} + \frac{Bs+C}{4s^2+20s+1} \\ &= \frac{15}{s} - \frac{60s+303}{4s^2+20s+1} + \frac{15}{s} \\ &= \frac{30}{s} - \frac{15s}{s^2+5s+0.25} + \frac{303}{4(s^2+5s+0.25)} \\ &= \frac{30}{s} - 15 \frac{\left(s+\frac{5}{2}\right)}{\left(s+\frac{5}{2}\right)^2-6} + \frac{75}{2} \frac{1}{\left(s+\frac{5}{2}\right)^2-6} + 75.75 \frac{1}{\left(s+\frac{5}{2}\right)^2-6} \end{aligned}$$

Calculate the inverse Laplace Transform.

$$\begin{aligned} v(t) &= 30u(t) - 15e^{-2.5t} \cosh(\sqrt{6}t) + \frac{37.5}{\sqrt{6}} e^{-2.5t} \sinh(\sqrt{6}t) + \frac{75.75}{\sqrt{6}} e^{-2.5t} \sinh(\sqrt{6}t) \\ &= 30u(t) - 15e^{-2.5t} \cosh(\sqrt{6}t) + 46.23e^{-2.5t} \sinh(\sqrt{6}t) \\ &= 30u(t) - 15e^{-2.5t} \left(\frac{e^{\sqrt{6}t} + e^{-\sqrt{6}t}}{2} \right) + 46.23e^{-2.5t} \left(\frac{e^{\sqrt{6}t} - e^{-\sqrt{6}t}}{2} \right) \\ &= 30u(t) - 7.5e^{-0.05t} - 7.5e^{-4.94t} + 23.12e^{-0.05t} - 23.12e^{-4.94t} \\ &= [30 + 15.62e^{-0.05t} - 30.62e^{-4.94t}] u(t) \text{ V} \end{aligned}$$

Therefore, the value of the voltage $v(t)$ is $\boxed{(30 + 15.62e^{-0.0501t} - 30.62e^{-4.949t})u(t) \text{ V}}$.